Easy Fence Brush

Official User Manual & Documentation

A non-destructive, path-based placement tool for Unity Level Designers.

Version 2.0

Contents

1 Introduction 3

2 GettingStarted 3
2.1 Creating Your First Fence Brush . . . .

3 DrawingthePath 4
3.1 SceneViewControls . . . .
3.2 PathOptions . . . .

4 ManagingAssets(VisualGridSystem) 4
4.1 ImportingPrefabs . . . .
4.2 VisualAssetSelection . . . .
4.3 WeightedProbabilitySystem (NEWinv2.0) . . . .
4.4 Per-PrefabAlwaysVertical (NEWinv2.0) . . . .

5 Posts&CornersSettings 6

6 Placement&Geometry 6

7 TerrainAdaptation
7.1 SlopeLimit (NEWinv2.0) 7

8 NaturalVariation&Damage 8

9 Finalizing:BakingtoPrefab 8

9 The Weight System 8

11 Tips & Best Practices 10

1 Introduction

Welcome to **Easy Fence Brush** — a non-destructive, path-based placement tool designed for Unity Level Designers and Environment Artists.
This tool eliminates the tedious process of manually placing and aligning repetitive objects. With an intuitive visual grid, smart terrain snapping, weighted randomization, and procedural variation, you can create complex, organic fences, walls, and barriers in seconds — directly in the Unity Scene View.

2 GettingStarted

2.1 Creating Your First Fence Brush

1. In the Unity Editor top menu, navigate to: **Tools > Easy Fence > Create Brush**
2. AnewGameObjectnamedFence Brushwillappearinyourscenehierarchy,automatically equipped with the EasyFenceTool component.
3. SelecttheFence Brushobjecttoopenthecustominspector.

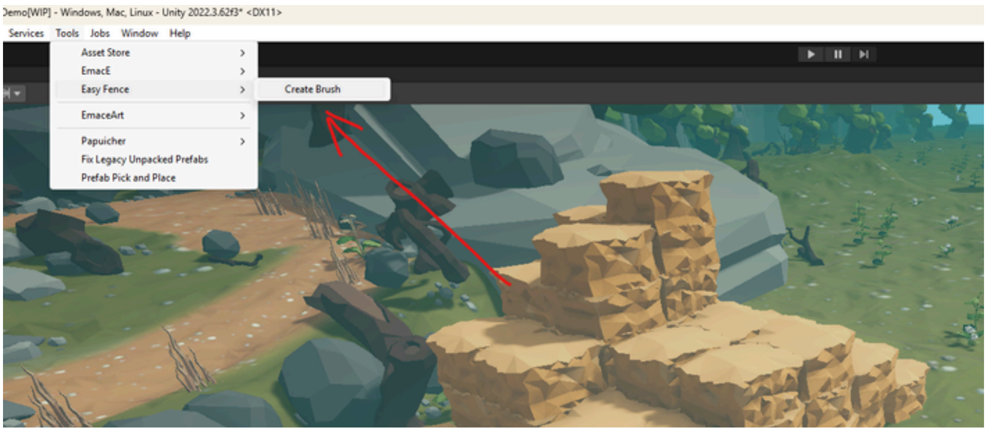

Figure 2: Accessing the tool via the top menu.

TIP: You can rename the GameObject in the hierarchy to anything you like — for example ”Garden Fence West” or ”Perimeter Wall A”. The baked output will inherit the name automatically.

3 DrawingthePath

The core of Easy Fence Brush is its non-destructive path system. To interact with the path in the Scene View, ensure **EDITING MODE** is active — the button in the inspector will be highlighted green.

If you are starting from scratch, click the � **ADD INITIAL POINT** � button to place the first node exactly where your Scene View camera is currently pointing.

Figure 3: The two primary control buttons in the inspector.

3.1 Scene View Controls

- **AddaPoint:** HoldShift + Left Clickontheterrainoranysurfacetoextendthefence path with a new node.
- **InsertaPoint:** HoldShift + Left Clickonthevirtuallinebetweentwoexistingpoints. The tool automatically splits the segment and inserts a new node at the closest position.
- **RemoveaPoint:** HoldCtrl + Left Clickonanexistingnode. ThenodeturnsREDto indicate it will be deleted on click.
- **Move a Point:** Click and drag the standard Unity Position Handle on any node to reposition it. The preview updates in real time.

NOTE: Make sure ”Gizmos” is enabled in your Scene View toolbar, otherwise the path handles will not be visible.

3.2 Path Options

- **Is Loop:** Connects the last point back to the first, creating a perfectly closed enclosure (e.g., a garden, a paddock).
- **Insert Threshold:** Controls how close your cursor must be to an existing segment to trigger an insert rather than appending to the end of the path. Lower values require more precision.

4 ManagingAssets(VisualGridSystem)

Easy Fence Brush separates your assets into two distinct sections:

- **FENCE SEGMENTS** — the repeating body panels of the fence.
- **POSTS & CORNERS** — the vertical pillars placed at path nodes.

Both sections work identically and support the full feature set described in this chapter.

4.1 Importing Prefabs

You have two modes, selectable via the toggle buttons at the top of each section:

- **Manual List:** Drag and drop individual prefabs from the Project window directly into the drop zone. Best for hand-picking a small, curated set of assets.

- **Folder Directory:** Drag and drop an entire folder from the Project window. The tool automatically finds and loads every prefab inside it, including subfolders. Best for large asset libraries.

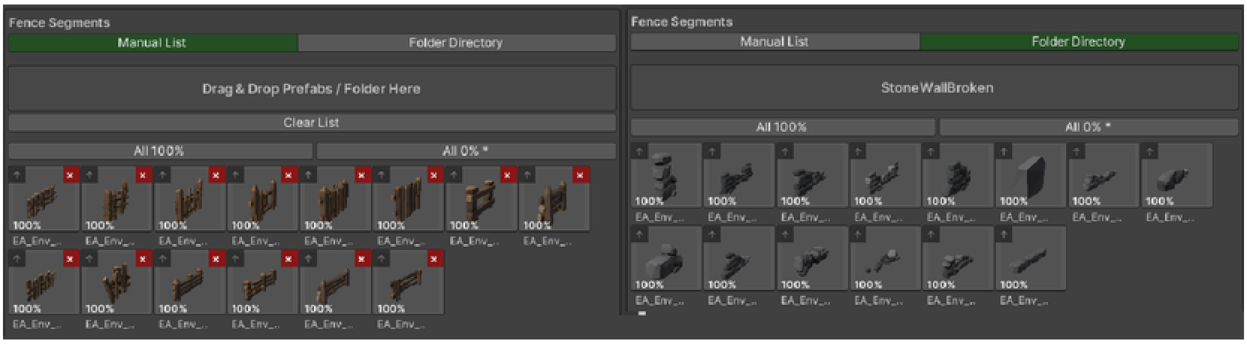

Figure 4: Importing assets into the brush.

4.2 Visual Asset Selection

Once imported, your prefabs appear as a grid of thumbnail images.

- **Include / Exclude:** Left-click any thumbnail to toggle it in or out of the generation pool. Excluded prefabs appear grayed out (30% opacity). This lets you test combinations without losing your asset references.

Note: Only left-click toggles include/exclude. Right-click on a thumbnail does nothing — this is intentional to prevent accidental changes.

4.3 WeightedProbabilitySystem(NEWinv2.0)

Every prefab in the grid has an individual spawn weight that controls how often it appears relative to the others.

- **Default Behavior:** All prefabs start at 100% weight — equal probability for every asset in the pool.

- **Adjusting Weight:** Click and drag **LEFT** on any thumbnail to reduce its weight. The thumbnail gradually darkens from the right edge as the weight decreases. The current weight percentage is always shown in the bottom-left corner of each thumbnail.
  - 100% = full brightness → prefab spawns at full rate
  - 50% = half darkened → prefab spawns half as often
  - 0% = fully dark → prefab never spawns

- **How Weights Work:** Weights are relative. If you have three prefabs at 100 / 50 / 100, the effective spawn chances are 40% / 20% / 40%.

Quick Buttons:

- **All 100%:** Resets all weights to 100% (equal distribution).
- **All0%*:** Setsallweightsto0%,exceptthefirstprefabwhichkeepsitslastsetvalue. Useful for quickly isolating one asset.

4.4 Per-Prefab Always Vertical (NEW in v2.0)

Each thumbnail has a small � icon in its top-left corner.

- **� (Dark, semi-transparent)** = default / follows global setting.
- **� (Bright Blue)** = this prefab is always vertical.

Clicking the � icon toggles the ”Always Vertical” override for that specific prefab. When active:

- The prefab always stands upright along the global Y-axis, regardless of terrain slope.
- Random Rotation Jitter on X and Z axes is suppressed for this prefab (Y-axis spin jitter is still applied).
- This overrides the global ”Keep Vertical” setting.

This is ideal for structural elements like fence posts or gate frames that must always appear vertical, even when used alongside panels that naturally follow the slope.

5 Posts & Corners Settings

Posts are the vertical pillars placed at path nodes. The tool offers intelligent placement logic so posts appear only where they make structural sense.

- **Placement Mode:** Choose from three sub-modes:
  - *All Points:* Places a post at every node on the path.
  - *Ends Only:* Places posts only at the very first and very last node.
  - *Corners Only (Smart Mode):* Places posts at both ends AND at any node where the path bends sharply.
- **Corner Angle Threshold:** Used in *Corners Only* mode. Sets the minimum bend angle (1° to 180°) required to spawn a post.
- **Post Vertical Offset:** Independently raises or lowers posts relative to the fence segments.

6 Placement & Geometry

Achieve perfect alignment without any mathematical guesswork.

- **AutoCalculateLength(Recommended: ON):**Whenenabled,thetoolscansthemesh bounds of each prefab to determine its exact physical length dynamically. Segments touch end-to-end perfectly with no gaps or overlaps.

- **Segment Length:** Manual override. Only active when Auto Calculate Length is disabled. The value is clamped to the physical minimum mesh size to prevent overlapping crashes.

- **Rotation Offset:** Fine-tunes the facing direction of segments. If your models were exported with an unusual forward axis, adjust the Y value here (e.g., Y=90).

- **Position Offset:** Shifts every segment relative to its placement position.

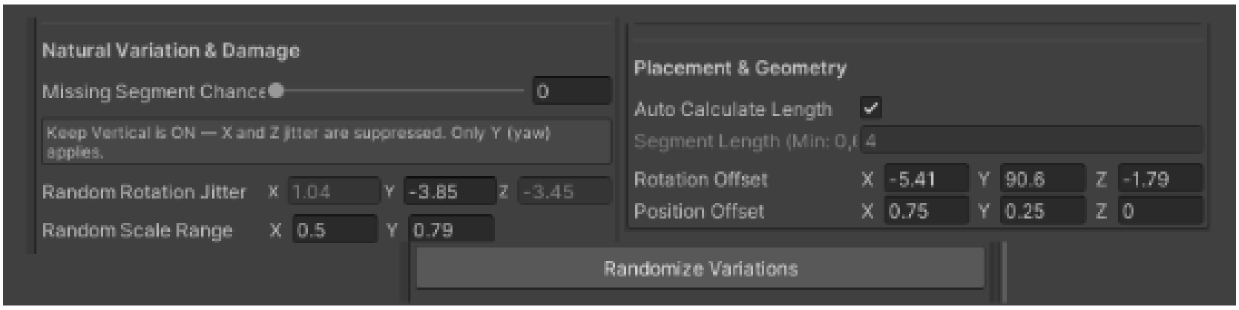

Figure 5: Geometry fine-tuning.

7 Terrain Adaptation

Make your fences look grounded and realistic regardless of the terrain topography.

- **Keep Vertical:** Forces ALL fence segments to stand straight up (parallel to the global Y-axis), even when climbing steep hills.

- **SnapToTerrain:** Projectseverypathpointandgeneratedobjectdownwardsviaraycast to rest perfectly on the surface below.

- **Snap Only To Unity Terrain:** Ignores regular 3D mesh colliders and only snaps to Unity Terrain components.

- **Terrain Mask:** The layer mask used for all snap raycasts.

- **Vertical Offset:** A global height adjustment applied to the entire fence (e.g., use a negative value to sink the base into the ground).

7.1 Slope Limit (NEW in v2.0)

- **Slope Limit (0° – 90°):** Sets the maximum surface angle (measured from horizontal) that the terrain snap system will accept. Surfaces steeper than this angle are ignored by the snap raycast.

Default is 90° (snaps to everything including vertical walls). Recommended for most fences is 45°–60°. If your fence unexpectedly jumps to a cliff edge, lower this limit.

8 NaturalVariation&Damage

Break up the repeating patterns of modular assets using procedural randomization to achieve a hand-crafted, organic look.

- **Missing Segment Chance (0 – 1):** Introduces random gaps. 0.2 means every segment has a 20% chance of not spawning. Great for ruined fences.
- **Random Rotation Jitter (X/Y/Z):** Applies a random angular rotation to each segment (Pitch/Yaw/Roll). Small values (5–15°) create a naturally weathered look.
- **RandomScaleRange(Min/Max):** Randomizesthesizeofeachsegment(e.g.,Min=0.9, Max=1.1).
- **� Randomize Variations:** Click this button to re-roll the random seed. The path stays unchanged, but all gaps, tilts, and scale variations generate a new unique pattern.

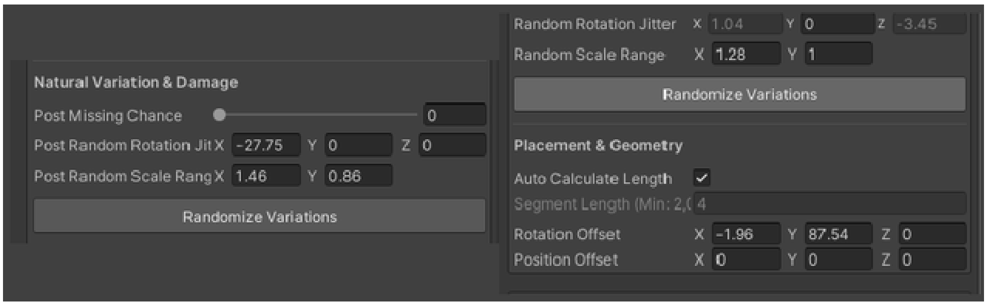

Figure 6: Re-roll the procedural noise seed.

9 Finalizing: Baking to Prefab

The fence visible during editing is a non-destructive Preview. Preview objects use special Unity flags to keep your hierarchy clean while you work.
When you are satisfied with the layout:

1. Scroll to the bottom of the inspector.
2. Click the green � **BAKE FENCE** � button.
3. A new clean GameObject (e.g. Fence Brush_Baked) appears in your hierarchy, containing all segments and posts as proper prefab instances.
4. You can now delete the original Fence Brush, or click **Clear Path** to draw a new fence.

10 The Weight System

1.1 Whattheweightsystemdoes

Every prefab in the grid has an individual **probability weight** expressed as a percentage (0–100%). When the tool places a fence segment or post, it picks a prefab from the active pool using weighted random selection — a prefab with weight 80% is roughly four times more likely to appear than one with weight 20%. This lets you control the visual frequency of each model without manually arranging them in a folder.

Weight 0% effectively removes a prefab from generation without deleting it from the list.

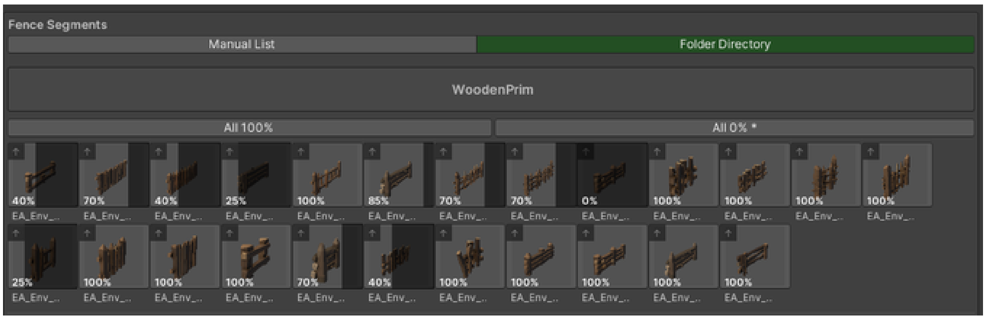

Figure 1: Visualizing probability distribution across the grid.

1.2 Reading the thumbnail

Each thumbnail displays its current weight in the bottom-left corner. A dark overlay grows from the right edge of the thumbnail as the weight decreases — at 50% half the image is darkened, at 0% the thumbnail is fully covered. This gives an instant visual read of the distribution across the entire grid without looking at numbers.

1.3 Adjustingweights

- **Drag to set** — click and drag horizontally on any thumbnail. Dragging right increases the weight, dragging left decreases it. The value steps in increments of 15 and is clamped between 0 and 100.
- **All 100%** — resets every prefab in the grid to equal weight.
- **All 0% *** — sets every prefab to 0% except the first one. Useful for quickly isolating a single model.

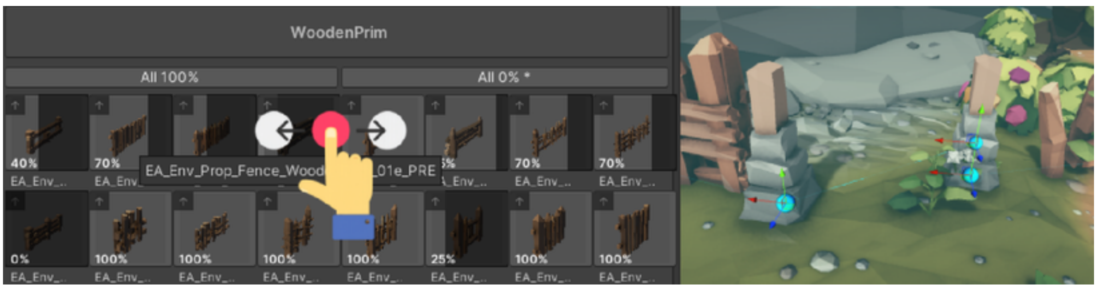

Figure 2: Adjusting the weight of a specific prefab.

1.4 Excluding a prefab

Left-clicking a thumbnail toggles that prefab between **active** and **excluded**. An excluded prefab is dimmed and takes no part in generation. Its weight is preserved and restored when you re-include it. The **Enable All** button appears whenever at least one prefab is excluded.

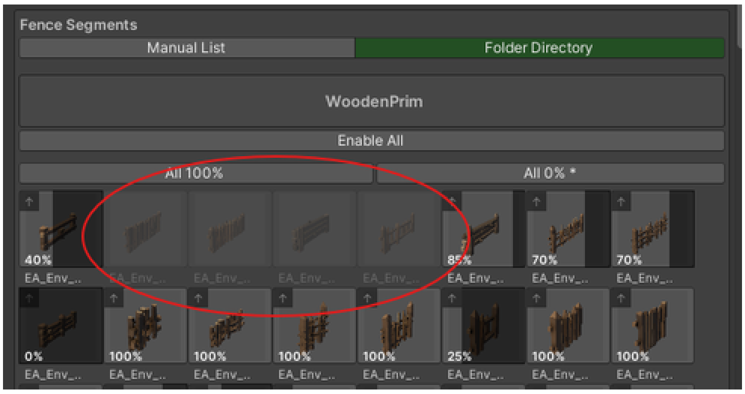

Figure 3: Excluding specific prefabs from the generation pool.

1.5 Removing a prefab

In Manual List mode a small [×] appears in the top-right corner of each thumbnail. Clicking it permanently removes that prefab from the list along with its stored weight. This is different from excluding — the prefab must be dragged back in to be used again. The × is not shown in Folder Directory mode.

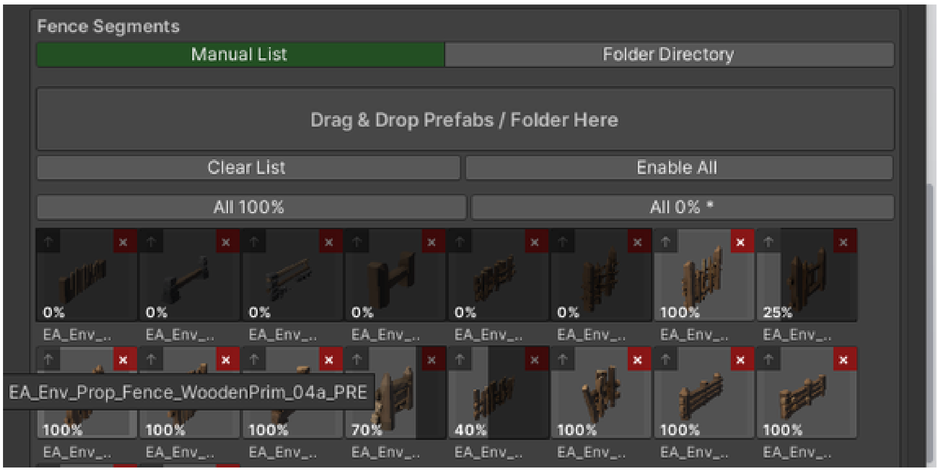

Figure 4: Removing a prefab from a manual list.

1.6 Tips

- Weights are relative, not absolute. A grid where all prefabs are at 50% behaves identically to one where all are at 100% — what matters is the ratio between values.
- Set rare damaged variants to 10–20% and common clean ones to 80–100% to create a naturally worn fence line with occasional broken sections.
- Use**All0%***thenmanuallyraiseindividualweightstobuildaprecisedistributionfrom scratch.

11 Tips & Best Practices

- **Multiple Fence Brushes:** You can have as many Fence Brush objects in your scene as you like. Each operates independently.
- **Workflow Order:** Draw path → Assign assets → Adjust weights → Set terrain options → Add variation → Bake.

Figure 7: Finalizing the fence into static geometry.

• **UndoSupport:** AlloperationssupportCtrl+Z / Cmd+Z,includingpointplacement,weight changes, and the Always Vertical flag.

• **Weight Strategy:** Set one "hero" prefab to 100% and supporting variants to 30–50% for a naturally dominant asset with occasional variety.

• **Performance:** For very long fences (500+ segments), consider splitting into multiple Fence Brush objects and baking each section separately for better static batching results.

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